

Managing SELinux TASK – 5

1. Check the current SELinux status on your Linux system using the `getenforce` and `sestatus` commands, and write down what each output means.

Ans :

Getenforce

Output :

Enforcing

Meaning of `getenforce` output:

- Enforcing – SELinux is active and enforcing security policies. Access violations are blocked.
- Permissive – SELinux is active but only logs violations; it does not block actions.
- Disabled – SELinux is completely turned off.

Sestatus

Output :

SELinux status	: enabled
SELinuxfs mount	: /sys/fs/selinux
SELinux root directory	: /etc/selinux
Loaded policy name	: targeted
Current mode	: enforcing
Mode from config file	: enforcing

Policy MLS status	: enabled
Policy deny_unknown status	: allowed
Memory protection checking	: actual (secure)
Max kernel policy version	: 33

Field	Meaning
SELinux status	Indicates whether SELinux is enabled or disabled.
SELinuxfs mount	Location where SELinux information is mounted in the filesystem.
SELinux root directory	Directory containing SELinux configuration and policy files.
Loaded policy name	The active policy type (commonly targeted, which protects specific services and processes).
Current mode	Current operating mode (enforcing, permissive, or disabled).

2. Switch SELinux to permissive mode temporarily using the setenforce command, then try creating a new file in /var/www/html and note any differences in error messages or permissions compared to enforcing mode.
Hint: Use setenforce 0 for permissive and setenforce 1 to return to enforcing.

Ans :

- Check the current SELinux mode
Getenforce
- Try creating a file in /var/www/html while SELinux is enforcing.
touch /var/www/html/testfile_enforcing
- Switch SELinux to permissive mode
sudo setenforce 0
- Try creating another file
touch /var/www/html/testfile_permissive
- Return SELinux to enforcing mode
sudo setenforce 1

**3. List all files in /var/www/html and use the ls -Z command to display their SELinux security contexts. Change the context of one file to httpd_sys_content_t using chcon and verify the change.

Hint: Use chcon --type=httpd_sys_content_t filename**

Ans :

- List files in /var/www/html
ls -l /var/www/html
- Display SELinux security contexts
ls -Z /var/www/html
- Change a file's context

```
sudo chcon --type=httpd_sys_content_t  
/var/www/html/testfile_permissive
```

- Verify the change

```
ls -Z /var/www/html/testfile_permissive
```

**4. A user is unable to upload images to a folder for their food review blog (like Zomato) hosted in /srv/foodie/images. Use the audit2why and audit2allow tools to analyze SELinux denial logs and suggest a solution to allow uploads.

Hint: Check /var/log/audit/audit.log for recent denials.**

Ans :

- Look for recent SELinux denials

```
sudo ausearch -m AVC -ts recent
```
- Analyze the denial with audit2why

```
sudo grep AVC /var/log/audit/audit.log | audit2why
```
- Generate a policy suggestion with audit2allow

```
sudo grep AVC /var/log/audit/audit.log | audit2allow
```
- Check the directory's SELinux context

```
ls -Zd /srv/foodie/images
```
- Relabel the upload directory

```
sudo semanage fcontext -a -t httpd_sys_rw_content_t  
"/srv/foodie/images(/.*)?"  
sudo restorecon -Rv /srv/foodie/images
```

